

Final Analysis Insight of Assignment 2

Project Title: PRODUCT INNOVATION HUB

Introduction: In this report Analyses Product Data set.

Key fields: *Product ID, Product Name, Brand Name, Price, Quantity, Category, Country Code, Manufacture date, Product Brand, Year, DD-MM-YY*

1. Handling Missing Values:

CLEAN FUNCTION: Removes line breaks, characters and other hidden control.

TRIM FUNCTION: This removes non printable characters and extra spaces at the same time.

ISBLANK FUNCTION: Checks whether cell is truly empty and returns True and False .

1) Check for missing values in the 'Price' column. How would you handle products with missing price information?

This means if the price is blank excel replaces it with average of the price column.

Syntax: =IF(ISBLANK(D2),AVERAGE(\$D\$2:\$D\$35),D2)

=IF(ISBLANK(D2),"Missing Value","Price Available")

Screenshot of Table:

Mode : Mode refers repetitive value.

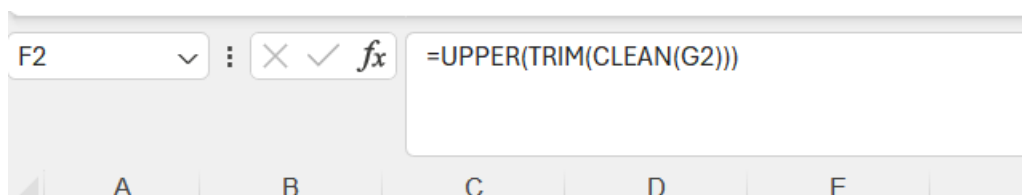
2) If there are products with missing categories, propose a strategy to impute or deal with these missing values effectively.

ISBLANK FUNCTION:

Syntax: =IF(ISBLANK(F13),"UNKNOWN",F13)

=IF(ISBLANK(D2),"Missing Category","Available Categories")

Screenshot of Categories Table:



Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category
28-JAN-US	Laptop	Dell	\$1,000	30	ELECTRONICS
15-FEB-US	Sneakers	Nike	\$80	15	FASHION
03-MAR-US	Coffee Maker	Keurig	\$130	40	KITCHEN
11-APR-US	Smartphone	Samsung	\$900	25	ELECTRONICS
22-MAY-US	Backpack	North Face	\$70	20	UNKNOWN
07-JUN-UK	Headphones	Sony	\$130	45	ELECTRONICS
19-JUL-UK	T-Shirt	Adidas	\$30	5	FASHION
23-AUG-UK	Blender	Ninja	\$90	35	KITCHEN
05-SEP-UK	Tablet	Apple	\$500	50	ELECTRONICS
14-OCT-UK	Hiking Boots	Timberland	\$130	10	OUTDOOR
17-JUN-IN	Laptop	Hp	\$950	25	ELECTRONICS
25-NOV-AU	Sneakers	Adidas	\$90	40	UNKNOWN
08-DEC-DE	Coffee Maker	Nespresso	\$120	35	UNKNOWN
18-FEB-CA	Smartwatch	Fitbit	\$150	15	ELECTRONICS
16-APR-ES	Headphones	Bose	\$250	20	ELECTRONICS
21-AUG-CA	Laptop Bag	Samsonite	\$50	35	ACCESSORIES
20-AUG-CN	Smartwatch	Huawei	\$160	15	ELECTRONICS
27-JAN-IT	Laptop	Asus	\$980	10	ELECTRONICS
01-MAR-UK	Sunglasses	Oakley	\$150	15	FASHION
14-AUG-US	Camping Tent	Coleman	\$130	10	OUTDOOR
14-MAY-RU	Camera	Nikon	\$700	50	ELECTRONICS
09-JAN-CA	Microwave	Panasonic	\$80	20	KITCHEN
19-JUL-BR	Fitness Tracker	Xiaomi	\$150	30	UNKNOWN

Answer: UNKNOWN

2. Correcting Inconsistent Data:

1) Identify any inconsistent text formats present in the "Product Name" column.

Syntax: =PROPER(TRIM(CLEAN(B2)))

Screenshot of Product Name Table:

The screenshot shows an Excel interface. The formula bar at the top displays the formula =PROPER(TRIM(CLEAN(C2))) for cell B2. Below it, a table with 5 columns (A to E) and 14 rows (1 to 14) is visible. The columns are labeled Product ID, Product Name, Brand Name, Price (\$), and Quantity. The data in the table is as follows:

	A	B	C	D	E
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity
2	28-JAN-US	Laptop	Dell	\$1,000	30
3	15-FEB-US	Sneakers	Nike	\$80	15
4	03-MAR-US	Coffee Maker	Keurig	\$130	40
5	11-APR-US	Smartphone	Samsung	\$900	25
6	22-MAY-US	Backpack	North Face	\$70	20
7	07-JUN-UK	Headphones	Sony	\$130	45
8	19-JUL-UK	T-Shirt	Adidas	\$30	5
9	23-AUG-UK	Blender	Ninja	\$90	35
10	05-SEP-UK	Tablet	Apple	\$500	50
11	14-OCT-UK	Hiking Boots	Timberland	\$130	10
12	17-JUN-IN	Laptop	Hp	\$950	25
13	25-NOV-AU	Sneakers	Adidas	\$90	40
14	08-DEC-DE	Coffee Maker	Nespresso	\$120	35

Answer: Laptop

2) Identify any typos present in the "Category" column.

Syntax: Find and Replace . Find "ELECTRONICSCS " to replace with "ELECTRONICS "

Screenshot of categories Table:

E	F
Quantity	Category
30	ELECTRONICS
15	FASHION
40	KITCHEN
25	ELECTRONICS
20	UNKNOWN
45	ELECTRONICS
5	FASHION
35	KITCHEN
50	ELECTRONICS
10	OUTDOOR
25	ELECTRONICS
40	UNKNOWN
35	UNKNOWN
15	ELECTRONICS
20	ELECTRONICS
35	ACCESSORIES
15	ELECTRONICS
10	ELECTRONICS
15	FASHION
10	OUTDOOR
50	ELECTRONICS
20	KITCHEN
30	UNKNOWN
45	ELECTRONICS
25	FASHION
40	KITCHEN

Answer: ELECTRONICS

3) Use the find and replace function to standardize the text formats in the "Product Name" column and fix any typos or misspellings in the "Category" column.

The screenshot shows a Microsoft Excel spreadsheet with the following data:

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category
28-JAN-US	Laptop	Dell	\$1,000	30	ELECTRONICS
15-FEB-US	Sneakers	Nike	\$80	15	FASHION
03-MAR-US	Coffee Maker	Keurig	\$130	40	KITCHEN
11-APR-US	Smartphone	Samsung	\$900	25	ELECTRONICS
22-MAY-US	Backpack	North Face	\$70	20	
07-JUN-UK	Headphones	Sony	\$130	45	ELECTRONICS
19-JUL-UK	T-Shirt	Adidas	\$30	5	FASHION
23-AUG-UK	Blender	Ninja	\$90	35	KITCHEN
05-SEP-UK	Tablet	Apple	\$500	50	ELECTRONICS
14-OCT-UK	Hiking Boots	Timberland	\$130	10	OUTDOOR
17-JUN-IN	Laptop	Hp	\$950	25	ELECTRONICS
25-NOV-AU	Sneakers	Adidas	\$90	40	
08-DEC-DE	Coffee Maker	Nespresso	\$120	35	

The "Find and Replace" dialog box is open, showing the message: "All done. We made 8 replacements." The "Find what:" field is empty, and the "Replace with:" field is also empty. The "Look in:" dropdown is set to "Formulas".

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category
28-JAN-US	Laptop	Dell	\$1,000	30	ELECTRONICS
15-FEB-US	Sneakers	Nike	\$80	15	FASHION
03-MAR-US	Coffee Maker	Keurig	\$130	40	KITCHEN
11-APR-US	Smartphone	Samsung	\$900	25	ELECTRONICS
22-MAY-US	Backpack	North Face	\$70	20	
07-JUN-UK	Headphones	Sony	\$130	45	ELECTRONICS
19-JUL-UK	T-Shirt	Adidas	\$30	5	FASHION
23-AUG-UK	Blender	Ninja	\$90	35	KITCHEN
05-SEP-UK	Tablet	Apple	\$500	50	ELECTRONICS
14-OCT-UK	Hiking Boots	Timberland	\$130	10	OUTDOOR
17-JUN-IN	Laptop	Hp	\$950	25	ELECTRONICS
25-NOV-AU	Sneakers	Adidas	\$90	40	
08-DEC-DE	Coffee Maker	Nespresso	\$120	35	
18-FEB-CA	Smartwatch	Fitbit	\$150	15	ELECTRONICS
16-APR-ES	Headphones	Bose	\$250	20	ELECTRONICS
21-AUG-CA	Laptop Bag	Samsonite	\$50	35	ACCESSORIES
20-AUG-CN	Smartwatch	Huawei	\$160	15	ELECTRONICS
27-JAN-IT	Laptop	Asus	\$980	10	ELECTRONICS
01-MAR-UK	Sunglasses	Oakley	\$150	15	FASHION
14-AUG-US	Camping Tent	Coleman	\$130	10	OUTDOOR
14-MAY-RU	Camera	Nikon	\$700	50	ELECTRONICS
09-JAN-CA	Microwave	Panasonic	\$80	20	KITCHEN
19-JUL-BR	Fitness Tracker	Xiaomi	\$150	30	
29-SEP-CA	Smartphone	Google	\$800	45	ELECTRONICS
03-JUN-CA	Sunglasses	Ray-Ban	\$130	25	FASHION
11-JUL-CA	Blender	Vitamix	\$400	40	KITCHEN
07-MAR-CA	Dress	Zara	\$60	30	FASHION

Create Table ? X

Where is the data for your table?

\$A\$1:\$F\$32

☒ My table has headers

OK Cancel

Dataset +

adv Accessibility: Investigate

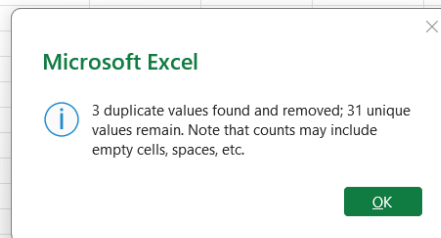
Final Table:

	A	B	C	D	E	F	G	H	I	J	K
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Country Code	Manufacture Date	Product Brand	DD-MM-YYYY	Year
2	28-JAN-US	Laptop	Dell	\$1,000	30	ELECTRONICS	US	28-JAN	Dell Laptop	28-JAN 2026	2026
3	15-FEB-US	Sneakers	Nike	\$80	15	FASHION	US	15-FEB	Nike Sneakers	15-FEB 2026	2026
4	03-MAR-US	Coffee Maker	Keurig	\$130	40	KITCHEN	US	03-MAR	Keurig Coffee Maker	03-MAR 2026	2026
5	11-APR-US	Smartphone	Samsung	\$900	25	ELECTRONICS	US	11-APR	Samsung Smartphone	11-APR 2026	2026
6	22-MAY-US	Backpack	North Face	\$70	20	UNKNOWN	US	22-MAY	North Face Backpack	22-MAY 2026	2026
7	07-JUN-UK	Headphones	Sony	\$130	45	ELECTRONICS	UK	07-JUN	Sony Headphones	07-JUN 2026	2026
8	19-JUL-UK	T-Shirt	Adidas	\$30	5	FASHION	UK	19-JUL	Adidas T-Shirt	19-JUL 2026	2026
9	23-AUG-UK	Blender	Ninja	\$90	35	KITCHEN	UK	23-AUG	Ninja Blender	23-AUG 2026	2026
10	05-SEP-UK	Tablet	Apple	\$500	50	ELECTRONICS	UK	05-SEP	Apple Tablet	05-SEP 2026	2026
11	14-OCT-UK	Hiking Boots	Timberland	\$130	10	OUTDOOR	UK	14-OCT	Timberland Hiking Boots	14-OCT 2026	2026
12	17-JUN-IN	Laptop	Hp	\$950	25	ELECTRONICS	IN	17-JUN	Hp Laptop	17-JUN 2026	2026
13	25-NOV-AU	Sneakers	Adidas	\$90	40	UNKNOWN	AU	25-NOV	Adidas Sneakers	25-NOV 2026	2026
14	08-DEC-DE	Coffee Maker	Nespresso	\$120	35	UNKNOWN	DE	08-DEC	Nespresso Coffee Maker	08-DEC 2026	2026
15	18-FEB-CA	Smartwatch	Fitbit	\$150	15	ELECTRONICS	CA	18-FEB	Fitbit Smartwatch	18-FEB 2026	2026
16	16-APR-ES	Headphones	Bose	\$250	20	ELECTRONICS	ES	16-APR	Bose Headphones	16-APR 2026	2026
17	21-AUG-CA	Laptop Bag	Samsonite	\$50	35	ACCESSORIES	CA	21-AUG	Samsonite Laptop Bag	21-AUG 2026	2026
18	20-AUG-CN	Smartwatch	Huawei	\$160	15	ELECTRONICS	CN	20-AUG	Huawei Smartwatch	20-AUG 2026	2026
19	27-JAN-IT	Laptop	Asus	\$980	10	ELECTRONICS	IT	27-JAN	Asus Laptop	27-JAN 2026	2026
20	01-MAR-UK	Sunglasses	Oakley	\$150	15	FASHION	UK	01-MAR	Oakley Sunglasses	01-MAR 2026	2026
21	14-AUG-US	Camping Tent	Coleman	\$130	10	OUTDOOR	US	14-AUG	Coleman Camping Tent	14-AUG 2026	2026
22	14-MAY-RU	Camera	Nikon	\$700	50	ELECTRONICS	RU	14-MAY	Nikon Camera	14-MAY 2026	2026
23	09-JAN-CA	Microwave	Panasonic	\$80	20	KITCHEN	CA	09-JAN	Panasonic Microwave	09-JAN 2026	2026
24	19-JUL-BR	Fitness Tracker	Xiaomi	\$150	30	UNKNOWN	BR	19-JUL	Xiaomi Fitness Tracker	19-JUL 2026	2026
25	29-SEP-CA	Smartphone	Google	\$800	45	ELECTRONICS	CA	29-SEP	Google Smartphone	29-SEP 2026	2026
26	03-JUN-CA	Sunglasses	Ray-Ban	\$130	25	FASHION	CA	03-JUN	Ray-Ban Sunglasses	03-JUN 2026	2026
27	11-JUL-CA	Blender	Vitamix	\$400	40	KITCHEN	CA	11-JUL	Vitamix Blender	11-JUL 2026	2026
28	07-MAR-CA	Dress	Zara	\$60	30	FASHION	CA	07-MAR	Zara Dress	07-MAR 2026	2026
29	13-APR-CA	Toaster	Hamilton	\$40	10	KITCHEN	CA	13-APR	Hamilton Toaster	13-APR 2026	2026
30	24-MAY-CA	Fitness Tracker	Garmin	\$130	5	ELECTRONICS	CA	24-MAY	Garmin Fitness Tracker	24-MAY 2026	2026
31	02-DEC-CA	Jeans	Levi'S	\$50	50	FASHION	CA	02-DEC	Levi'S Jeans	02-DEC 2026	2026
32	09-JUL-FR	Watch	Casio	\$100	20	ACCESSORIES	FR	09-JUL	Casio Watch	09-JUL 2026	2026

➤ Removing Duplicates:

1) Identify any duplicate rows within the dataset based on the entirety of each row, and remove them if any.

	A	B	C	D	E	F	G	H	I	J	K
	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category					
	28-JAN-US	laptop	Dell	1000	30	Electronics					
	15-FEB-US	Sneakers	Nike	80	15	Fashion					
	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen					
	11-APR-US	smartphone	Samsung	900	25	Electronics					
	22-MAY-US	Backpack	North Face	70	20						
	07-JUN-UK	Headphones	Sony		45	Electroni					
	19-JUL-UK	T-shirt	Adidas	30	5	Fashion					
	23-AUG-UK	Blender	Ninja	90	35	Kitchen					
	05-SEP-UK	Tablet	Apple	500	50	Electronics					
	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor					
	17-JUN-IN	Laptop	HP	950	25	Electronics					
	25-NOV-AU	Sneakers	Adidas	90	40						
	08-DEC-DE	Coffee Maker	Nespresso	120	35						
	18-FEB-CA	Smartwatch	Fitbit	150	15	Electroni					
	16-APR-ES	headphones	Bose	250	20	Electroni					
	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories					
	20-AUG-CN	Smartwatch	Huawei	160	15	Electroni					
	27-JAN-IT	laptop	Asus	980	10	Electronics					
	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion					
	14-AUG-US	Camping Tent	Coleman		10	Outdoor					
	14-MAY-RU	Camera	Nikon	700	50	Electronics					
	09-JAN-CA	Microwave	Panasonic	80	20	Kitchen					
	19-JUL-BR	Fitness Tracker	Xiaomi	150	30						
	29-SEP-CA	Smartphone	Google	800	45	Electronics					
	03-JUN-CA	Sunglasses	Ray-Ban		25	Fashion					
	11-JUL-CA	Blender	Vitamix	400	40	Kitchen					
	07-MAR-CA	Dress	Zara	60	30	Fashion					



The below listed 3 row is Duplicate row in the Dataset.

Answers:

12	17-JUN-IN	Laptop	HP	950	25	Electronics
	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
16	16-APR-ES	headphones	Bose	250	20	Electroni

4. Splitting and Merging Data:

1) Split the "Product ID" column into two separate columns for " Manufacturing Date" and "Country Code". Remove unnecessary characters, if any.

- Country Code:

Syntax: =RIGHT(A2,2)

Screenshot of Country Code Table:

N50 $\times$ $\checkmark$ f_x =RIGHT(A2,2)

	A	B	C	D	E	F	G	H	I	J	K
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Country Code	Manufacture Date	Product Brand	DD-MM-YYYY	Year
2	28-JAN-US	Laptop	Dell	\$1,000	30	ELECTRONICS	US	28-JAN	Dell Laptop	28-JAN 2026	2026
3	15-FEB-US	Sneakers	Nike	\$80	15	FASHION	US	15-FEB	Nike Sneakers	15-FEB 2026	2026
4	03-MAR-US	Coffee Maker	Keurig	\$130	40	KITCHEN	US	03-MAR	Keurig Coffee Maker	03-MAR 2026	2026
5	11-APR-US	Smartphone	Samsung	\$900	25	ELECTRONICS	US	11-APR	Samsung Smartphone	11-APR 2026	2026
6	22-MAY-US	Backpack	North Face	\$70	20	UNKNOWN	US	22-MAY	North Face Backpack	22-MAY 2026	2026
7	07-JUN-UK	Headphones	Sony	\$130	45	ELECTRONICS	UK	07-JUN	Sony Headphones	07-JUN 2026	2026
8	19-JUL-UK	T-Shirt	Adidas	\$30	5	FASHION	UK	19-JUL	Adidas T-Shirt	19-JUL 2026	2026
9	23-AUG-UK	Blender	Ninja	\$90	35	KITCHEN	UK	23-AUG	Ninja Blender	23-AUG 2026	2026
10	05-SEP-UK	Tablet	Apple	\$500	50	ELECTRONICS	UK	05-SEP	Apple Tablet	05-SEP 2026	2026
11	14-OCT-UK	Hiking Boots	Timberland	\$130	10	OUTDOOR	UK	14-OCT	Timberland Hiking Boots	14-OCT 2026	2026
12	17-JUN-IN	Laptop	Hp	\$950	25	ELECTRONICS	IN	17-JUN	Hp Laptop	17-JUN 2026	2026
13	25-NOV-AU	Sneakers	Adidas	\$90	40	UNKNOWN	AU	25-NOV	Adidas Sneakers	25-NOV 2026	2026
14	08-DEC-DE	Coffee Maker	Nespresso	\$120	35	UNKNOWN	DE	08-DEC	Nespresso Coffee Maker	08-DEC 2026	2026
15	18-FEB-CA	Smartwatch	Fitbit	\$150	15	ELECTRONICS	CA	18-FEB	Fitbit Smartwatch	18-FEB 2026	2026
16	16-APR-ES	Headphones	Bose	\$250	20	ELECTRONICS	ES	16-APR	Bose Headphones	16-APR 2026	2026

Answer: **US**

- **Manufacture Date:**

Syntax: =LEFT(A2,6)

Screenshot of Manufacture DateTable:

N51 $\times$ $\checkmark$ f_x =LEFT(A2,6)

	A	B	C	D	E	F	G	H	I	J	K
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Country Code	Manufacture Date	Product Brand	DD-MM-YYYY	Year
2	28-JAN-US	Laptop	Dell	\$1,000	30	ELECTRONICS	US	28-JAN	Dell Laptop	28-JAN 2026	2026
3	15-FEB-US	Sneakers	Nike	\$80	15	FASHION	US	15-FEB	Nike Sneakers	15-FEB 2026	2026
4	03-MAR-US	Coffee Maker	Keurig	\$130	40	KITCHEN	US	03-MAR	Keurig Coffee Maker	03-MAR 2026	2026
5	11-APR-US	Smartphone	Samsung	\$900	25	ELECTRONICS	US	11-APR	Samsung Smartphone	11-APR 2026	2026
6	22-MAY-US	Backpack	North Face	\$70	20	UNKNOWN	US	22-MAY	North Face Backpack	22-MAY 2026	2026
7	07-JUN-UK	Headphones	Sony	\$130	45	ELECTRONICS	UK	07-JUN	Sony Headphones	07-JUN 2026	2026
8	19-JUL-UK	T-Shirt	Adidas	\$30	5	FASHION	UK	19-JUL	Adidas T-Shirt	19-JUL 2026	2026
9	23-AUG-UK	Blender	Ninja	\$90	35	KITCHEN	UK	23-AUG	Ninja Blender	23-AUG 2026	2026
10	05-SEP-UK	Tablet	Apple	\$500	50	ELECTRONICS	UK	05-SEP	Apple Tablet	05-SEP 2026	2026
11	14-OCT-UK	Hiking Boots	Timberland	\$130	10	OUTDOOR	UK	14-OCT	Timberland Hiking Boots	14-OCT 2026	2026
12	17-JUN-IN	Laptop	Hp	\$950	25	ELECTRONICS	IN	17-JUN	Hp Laptop	17-JUN 2026	2026
13	25-NOV-AU	Sneakers	Adidas	\$90	40	UNKNOWN	AU	25-NOV	Adidas Sneakers	25-NOV 2026	2026
14	08-DEC-DE	Coffee Maker	Nespresso	\$120	35	UNKNOWN	DE	08-DEC	Nespresso Coffee Maker	08-DEC 2026	2026
15	18-FEB-CA	Smartwatch	Fitbit	\$150	15	ELECTRONICS	CA	18-FEB	Fitbit Smartwatch	18-FEB 2026	2026
16	16-APR-ES	Headphones	Bose	\$250	20	ELECTRONICS	ES	16-APR	Bose Headphones	16-APR 2026	2026
17	21-AUG-CA	Laptop Bag	Samsonite	\$50	35	ACCESSORIES	CA	21-AUG	Samsonite Laptop Bag	21-AUG 2026	2026
18	20-AUG-CN	Smartwatch	Huawei	\$160	15	ELECTRONICS	CN	20-AUG	Huawei Smartwatch	20-AUG 2026	2026
19	27-JAN-IT	Laptop	Asus	\$980	10	ELECTRONICS	IT	27-JAN	Asus Laptop	27-JAN 2026	2026

Answer: **28-JAN**

2) Merge the "Brand Name" and "Product Name" columns into one column named "Product Brand".

- **Product Brand:**

Syntax: =CONCATENATE(C2," ",B2)

Screenshot of Product Brand DateTable:

N57 =CONCATENATE(C2," ",B2)

	A	B	C	D	E	F	G	H	I	J	K
	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Country Code	Manufacture Date	Product Brand	DD-MM-YYYY	Year
1	28-JAN-US	Laptop	Dell	\$1,000	30	ELECTRONICS	US	28-JAN	Dell Laptop	28-JAN 2026	2026
2	15-FEB-US	Sneakers	Nike	\$80	15	FASHION	US	15-FEB	Nike Sneakers	15-FEB 2026	2026
3	03-MAR-US	Coffee Maker	Keurig	\$130	40	KITCHEN	US	03-MAR	Keurig Coffee Maker	03-MAR 2026	2026
4	11-APR-US	Smartphone	Samsung	\$900	25	ELECTRONICS	US	11-APR	Samsung Smartphone	11-APR 2026	2026
5	22-MAY-US	Backpack	North Face	\$70	20	UNKNOWN	US	22-MAY	North Face Backpack	22-MAY 2026	2026
6	07-JUN-UK	Headphones	Sony	\$130	45	ELECTRONICS	UK	07-JUN	Sony Headphones	07-JUN 2026	2026
7	19-JUL-UK	T-Shirt	Adidas	\$30	5	FASHION	UK	19-JUL	Adidas T-Shirt	19-JUL 2026	2026
8	23-AUG-UK	Blender	Ninja	\$90	35	KITCHEN	UK	23-AUG	Ninja Blender	23-AUG 2026	2026
9	05-SEP-UK	Tablet	Apple	\$500	50	ELECTRONICS	UK	05-SEP	Apple Tablet	05-SEP 2026	2026
0	14-OCT-UK	Hiking Boots	Timberland	\$130	10	OUTDOOR	UK	14-OCT	Timberland Hiking Boots	14-OCT 2026	2026
1	17-JUN-IN	Laptop	Hp	\$950	25	ELECTRONICS	IN	17-JUN	Hp Laptop	17-JUN 2026	2026
2	25-NOV-AU	Sneakers	Adidas	\$90	40	UNKNOWN	AU	25-NOV	Adidas Sneakers	25-NOV 2026	2026
3	08-DEC-DE	Coffee Maker	Nespresso	\$120	35	UNKNOWN	DE	08-DEC	Nespresso Coffee Maker	08-DEC 2026	2026
4	18-FEB-CA	Smartwatch	Fitbit	\$150	15	ELECTRONICS	CA	18-FEB	Fitbit Smartwatch	18-FEB 2026	2026
5	15-APR-ES	Headphones	Sony	\$130	45	ELECTRONICS	ES	15-APR	Sony Headphones	15-APR 2026	2026

Answer: **Dell Laptop**

5. Number Formatting:

1)Format the data type of the "Price" column to a currency format.

- **Price:**

Syntax: GOTO Home page-right click format cells -select category -Currency
Format-select unites states

Screenshot of Price\$ Table:

D
Price (\$)
\$1,000
\$80
\$130
\$900
\$70
\$130
\$30
\$90
\$500
\$130
\$950
\$90
\$120
\$150
\$250
\$50
\$160

File

Home

Insert

Draw

Page Layout

Formulas

Data

Review

View

Help

Acrobat

Table Design

Paste

Clipboard

Font

Alignment

Number

D2

1000

	A	B	C	D	E	F	G	H	I
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Country Code	Manufacture Date	Product Brand
2	28-JAN-US	Laptop	Dell	\$1,000	30	ELECTRONICS	US	28-JAN	Dell Laptop
3	15-FEB-US	Sneakers	Nike	\$80	15	FASHION	US	15-FEB	Nike Sneakers
4	03-MAR-US	Coffee Maker	Keurig	\$130	40	KITCHEN	US	03-MAR	Keurig Coffee Maker

Answer: **\$1000**

2) Format the "Manufacturing Date" column to display dates in the "DD-MM-YYYY " format.

Syntax: **=YEAR(H2)**

=CONCATENATE(H2," ",N71)

Screenshot of DD-MM-YYYY Table:

=YEAR(H2)

C	D	E	F	G	H	I	J	K
Brand Name	Price (\$)	Quantity	Category	Country Code	Manufacture Date	Product Brand	DD-MM-YYYY	Year
Dell	\$1,000	30	ELECTRONICS	US	28-JAN	Dell Laptop	28-JAN 2026	2026
Nike	\$80	15	FASHION	US	15-FEB	Nike Sneakers	15-FEB 2026	2026
Keurig	\$130	40	KITCHEN	US	03-MAR	Keurig Coffee Maker	03-MAR 2026	2026
Samsung	\$900	25	ELECTRONICS	US	11-APR	Samsung Smartphone	11-APR 2026	2026
North Face	\$70	20	UNKNOWN	US	22-MAY	North Face Backpack	22-MAY 2026	2026
Sony	\$130	45	ELECTRONICS	UK	07-JUN	Sony Headphones	07-JUN 2026	2026
Adidas	\$30	5	FASHION	UK	19-JUL	Adidas T-Shirt	19-JUL 2026	2026
Ninja	\$90	35	KITCHEN	UK	23-AUG	Ninja Blender	23-AUG 2026	2026
Apple	\$500	50	ELECTRONICS	UK	05-SEP	Apple Tablet	05-SEP 2026	2026
Timberland	\$130	10	OUTDOOR	UK	14-OCT	Timberland Hiking Boots	14-OCT 2026	2026
Hp	\$950	25	ELECTRONICS	IN	17-JUN	Hp Laptop	17-JUN 2026	2026
Adidas	\$90	40	UNKNOWN	AU	25-NOV	Adidas Sneakers	25-NOV 2026	2026
Nespresso	\$120	35	UNKNOWN	DE	08-DEC	Nespresso Coffee Maker	08-DEC 2026	2026
Fitbit	\$150	15	ELECTRONICS	CA	18-FEB	Fitbit Smartwatch	18-FEB 2026	2026
Bose	\$250	20	ELECTRONICS	ES	16-APR	Bose Headphones	16-APR 2026	2026
Samsonite	\$50	35	ACCESSORIES	CA	21-AUG	Samsonite Laptop Bag	21-AUG 2026	2026
Huawei	\$160	15	ELECTRONICS	CN	20-AUG	Huawei Smartwatch	20-AUG 2026	2026
Asus	\$980	10	ELECTRONICS	IT	27-JAN	Asus Laptop	27-JAN 2026	2026
Oakley	\$150	15	FASHION	UK	01-MAR	Oakley Sunglasses	01-MAR 2026	2026
Coleman	\$120	10	OUTDOOR	US	14-AUG	Coleman Camping Tent	14-AUG 2026	2026

N72 =CONCATENATE(H2,"",N71)

A	B	C	D	E	F	G	H	I	J	K
Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Country Code	Manufacture Date	Product Brand	DD-MM-YYYY	Year
28-JAN-US	Laptop	Dell	\$1,000	30	ELECTRONICS	US	28-JAN	Dell Laptop	28-JAN 2026	2026
15-FEB-US	Sneakers	Nike	\$80	15	FASHION	US	15-FEB	Nike Sneakers	15-FEB 2026	2026
03-MAR-US	Coffee Maker	Keurig	\$130	40	KITCHEN	US	03-MAR	Keurig Coffee Maker	03-MAR 2026	2026
11-APR-US	Smartphone	Samsung	\$900	25	ELECTRONICS	US	11-APR	Samsung Smartphone	11-APR 2026	2026
22-MAY-US	Backpack	North Face	\$70	20	UNKNOWN	US	22-MAY	North Face Backpack	22-MAY 2026	2026
07-JUN-UK	Headphones	Sony	\$130	45	ELECTRONICS	UK	07-JUN	Sony Headphones	07-JUN 2026	2026
19-JUL-UK	T-Shirt	Adidas	\$30	5	FASHION	UK	19-JUL	Adidas T-Shirt	19-JUL 2026	2026
23-AUG-UK	Blender	Ninja	\$90	35	KITCHEN	UK	23-AUG	Ninja Blender	23-AUG 2026	2026

Answer: 28-01-2026

6. Conditional Formatting:

1) Apply data bar or color scales conditional formatting in the "Price" column.

Syntax: In Home tab we have conditional formatting, click arrow, Select Data Bar.

D
Price (\$)
\$1,000
\$80
\$130
\$900
\$70
\$130
\$30
\$90
\$500
\$130
\$950
\$90
\$120
\$150
\$250
\$50

Screenshot of PriceTable:

2) Create a custom rule for conditional formatting in the "Category" column to highlight cells where the category is "Electronics."

Syntax: =\$F2="ELECTRONICS"

N86						
A	B	C	D	E	F	
Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	
28-JAN-US	Laptop	Dell	\$1,000	30	ELECTRONICS	
15-FEB-US	Sneakers	Nike	\$80	15	FASHION	
03-MAR-US	Coffee Maker	Keurig	\$130	40	KITCHEN	
11-APR-US	Smartphone	Samsung	\$900	25	ELECTRONICS	
22-MAY-US	Backpack	North Face	\$70	20	UNKNOWN	
07-JUN-UK	Headphones	Sony	\$130	45	ELECTRONICS	
19-JUL-UK	T-Shirt	Adidas	\$30	5	FASHION	
23-AUG-UK	Blender	Ninja	\$90	35	KITCHEN	
05-SEP-UK	Tablet	Apple	\$500	50	ELECTRONICS	

Category	C
ELECTRONICS	
FASHION	
KITCHEN	
ELECTRONICS	
UNKNOWN	
ELECTRONICS	
FASHION	
KITCHEN	
ELECTRONICS	
OUTDOOR	
ELECTRONICS	

Screenshot of Category Table:

Answer:

ELECTRONICS

Conclusion:

In this data exploration. assessed the structure and quality of data set and identified missing values and remove duplicates. The analysis revealed clear pattern. Clear chart of visualization.